Signali i sustavi

LV6 – Moj Shazam – Prvi dio

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Preddiplomski sveučilišni studij računarstva, programsko inženjerstvo

PR-P-LV1/2

Na ovoj vježbi napisat ćete skriptu **make_table** koja će primati pjesmu i konstruirati bazu značajki za predanu pjesmu. Skripta će raditi sljedeće:
1. Učitati pjesmu korištenjem predane **audioread** funkcije.
2. Usrednjiti dva kanala, oduzeti prosječnu vrijednost pjesme te poduzorkovati (*downsample*).
3. Izračunati (log magnitude) spektogram pjesme koristeći naredbu spectogram.
4. Pronaći lokalne vrhove spektograma korištenjem **circshift** naredbe u petlji.
5. Adaptivno odrediti prag rezultata iz koraka 4 kako bi imali **peak_rate** vrhova u sekundi.
6. Za svaki vrh pronađite i odaberite **fanout** parova unutar dozvoljenog prozora i dodajte ih u tablicu.
7. Plotajte mapu konstelacije (*constellation map*) i iscrtajte linije koje spajaju odabrane parove.

Kod:

```
[y,fs]= audioread('tonev2018.mp3');
y=mean(y,2);
y=y-mean(y);
FS=8000;
y_resample=resample(y , FS , fs);
[S,F,T]=spectrogram(y_resample,512,256,512,FS);
imagesc(F,T,real(S));
title('Magnituda')
xlabel('Frekvencija')
ylabel('Vrijeme')
figure
imagesc(F,T,real(log(S)));
title('Log magnituda')
xlabel('Frekvencija' )
ylabel('Vrijeme')
S=real(log(S));
gs=9;
P=zeros('like',S);
for i=int64(-gs/2):1:int64(gs/2)
for j=int64(-gs/2):1:int64(gs/2)
CS=circshift(S,[j,i]);
P=P+((S-CS)>0);
end
end
P=(P-max(P(:))>=0);
figure
imagesc(T,F,real(P));
colormap(1-gray);
title('Mapa konstelacije')
ylabel('Frekvencija')
xlabel('Vrijeme')
```

Grafovi:

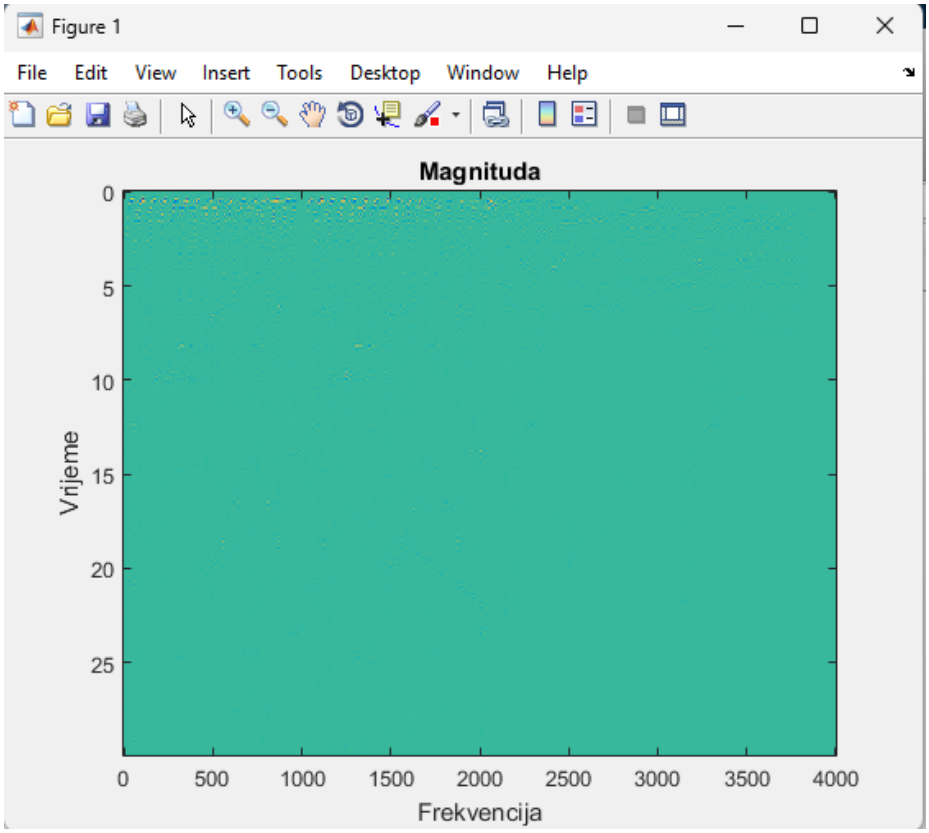
Figure 1
File Edit View Insert Tools Desktop Window Help
Magnituda
Vrijeme
0
5
10
15
20
25
0 500 1000 1500 2000 2500 3000 3500 4000
Frekvencija

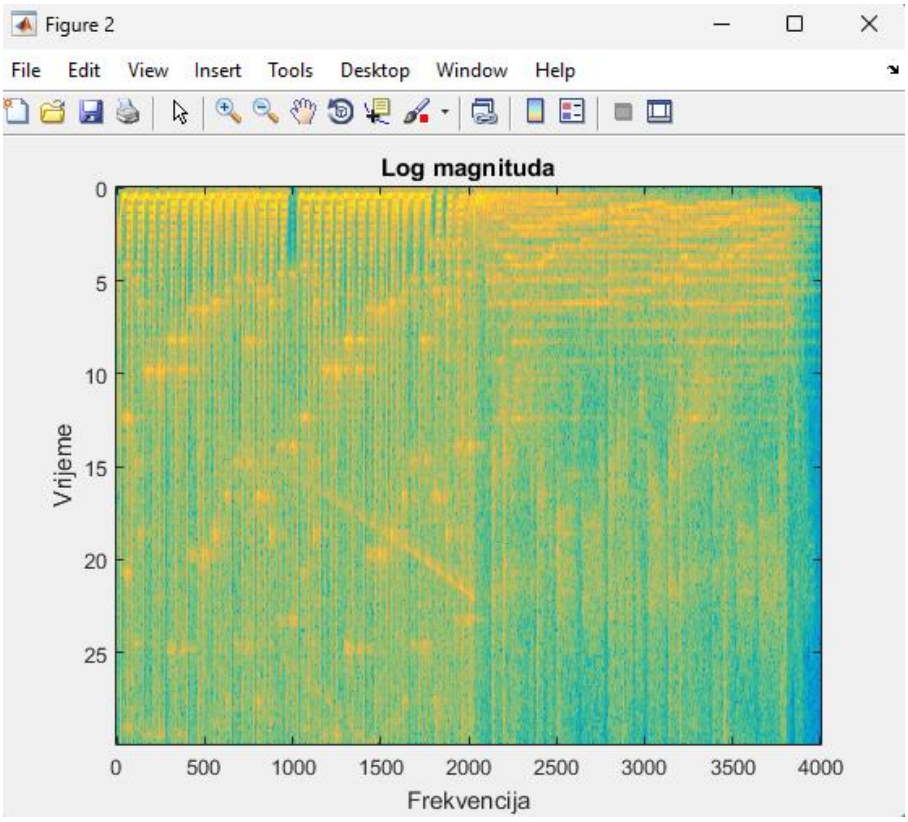
Figure 2
File Edit View Insert Tools Desktop Window Help
Log magnituda
Vrijeme
0
5
10
15
20
25
0 500 1000 1500 2000 2500 3000 3500 4000
Frekvencija

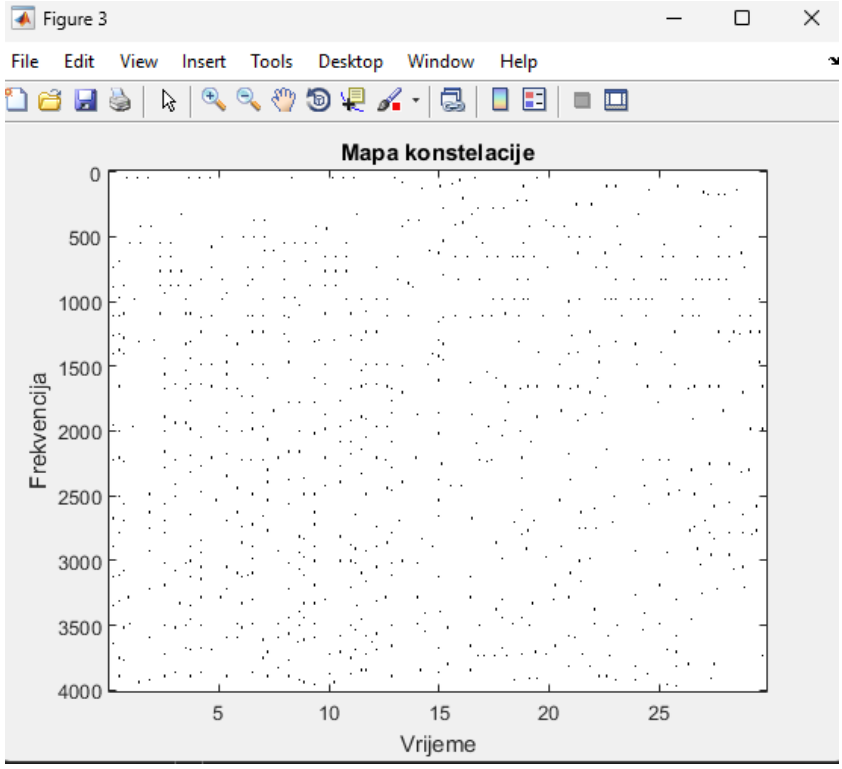
Figure 3
File Edit View Insert Tools Desktop Window Help
Mapa konstelacije
Frekvencija
0
500
1000
1500
2000
2500
3000
3500
4000
5
10
15
20
25
Vrijeme

Za gs=5

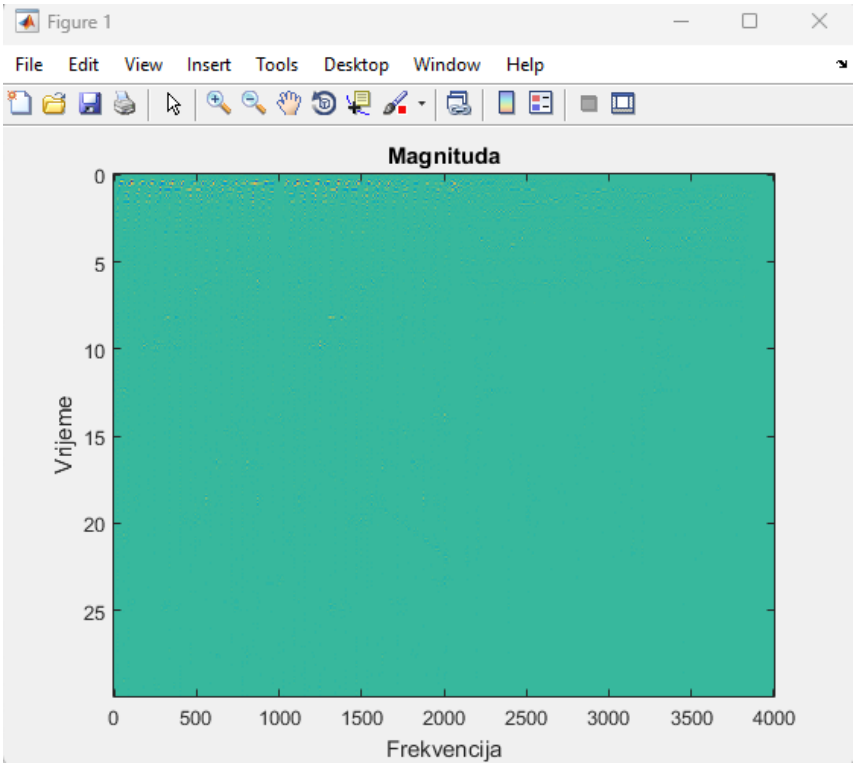
Figure 1
File Edit View Insert Tools Desktop Window Help
Magnituda
Vrijeme
0
5
10
15
20
25
0
500
1000
1500
2000
2500
3000
3500
4000
Frekvencija

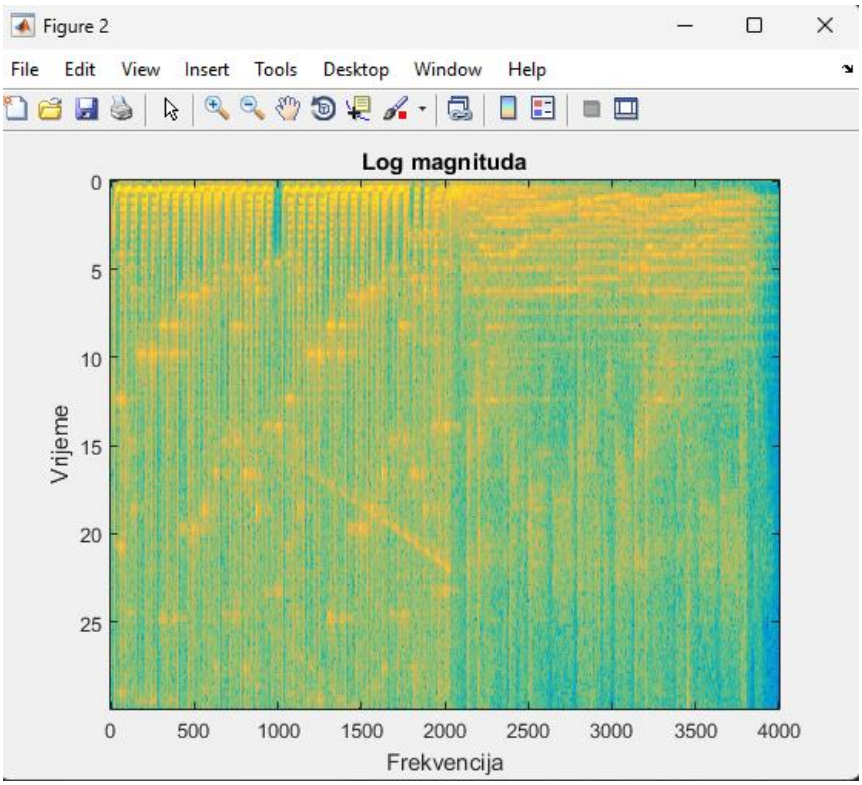
Figure 2
File Edit View Insert Tools Desktop Window Help
Log magnituda
Vrijeme
0
5
10
15
20
25
0 500 1000 1500 2000 2500 3000 3500 4000
Frekvencija

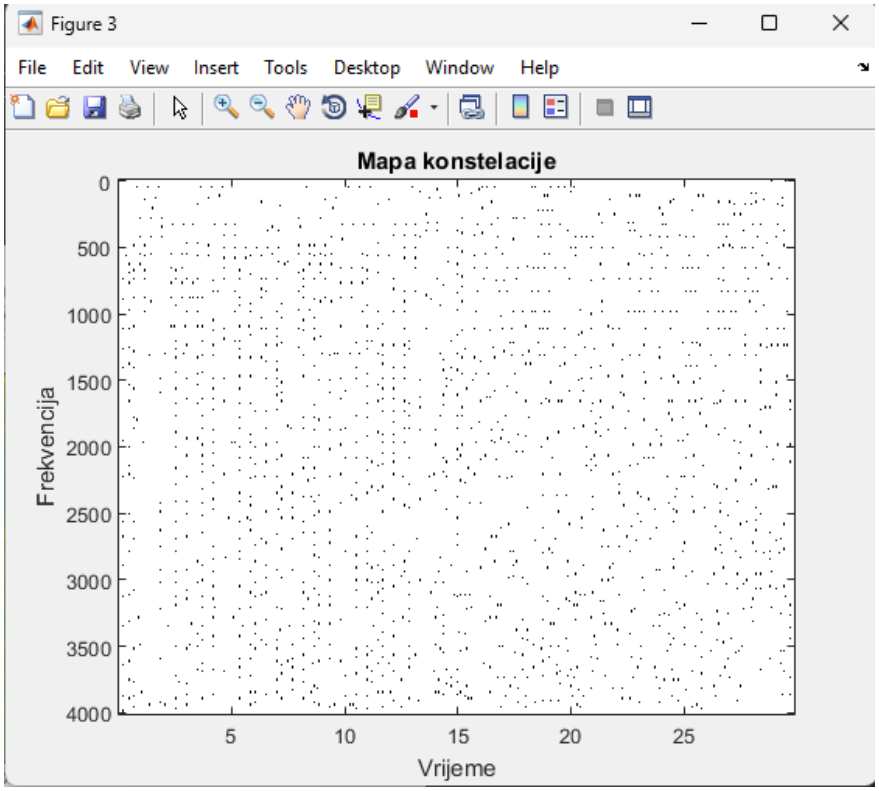
Figure 3
File Edit View Insert Tools Desktop Window Help
Mapa konstelacije
Frekvencija
0
500
1000
1500
2000
2500
3000
3500
4000
5 10 15 20 25
Vrijeme

Kod:

```
function t=threshold_peaks(S,T,P,nP)
t=[];
threshold=length(T) / T(end);
chunk_start=1;
while chunk_start <length(T)
if chunk_start+floor(threshold) >length(T)
chunk_kraj=length(T);
else
chunk_kraj=chunk_start+floor(threshold);
end
chunck_vrijeme=T(chunk_start:chunk_kraj);
chunck_nP=round(nP*(chunck_vrijeme(end)-chunck_vrijeme(1)));
if chunck_nP>0
chunk_s=s(:,chunk_start:chunk_kraj);
chunk_p=p(:,chunk_start:chunk_kraj);
chunk_ps=chunk_s(chunk_p);
sorted_ps=sort(reshape(chunk_ps.',1,[]),'descend');
if length(sorted_ps) > 0 && chunk_nP <= numel(sorted_ps)
chunk_treshold=sorted_ps(chunk_nP);
chunk_p(chunk_s< chunk_treshold)=0;
t=[t,chunk_p];
end
end
chunk_start=chunk_kraj+1;
end
end
```

Graf:

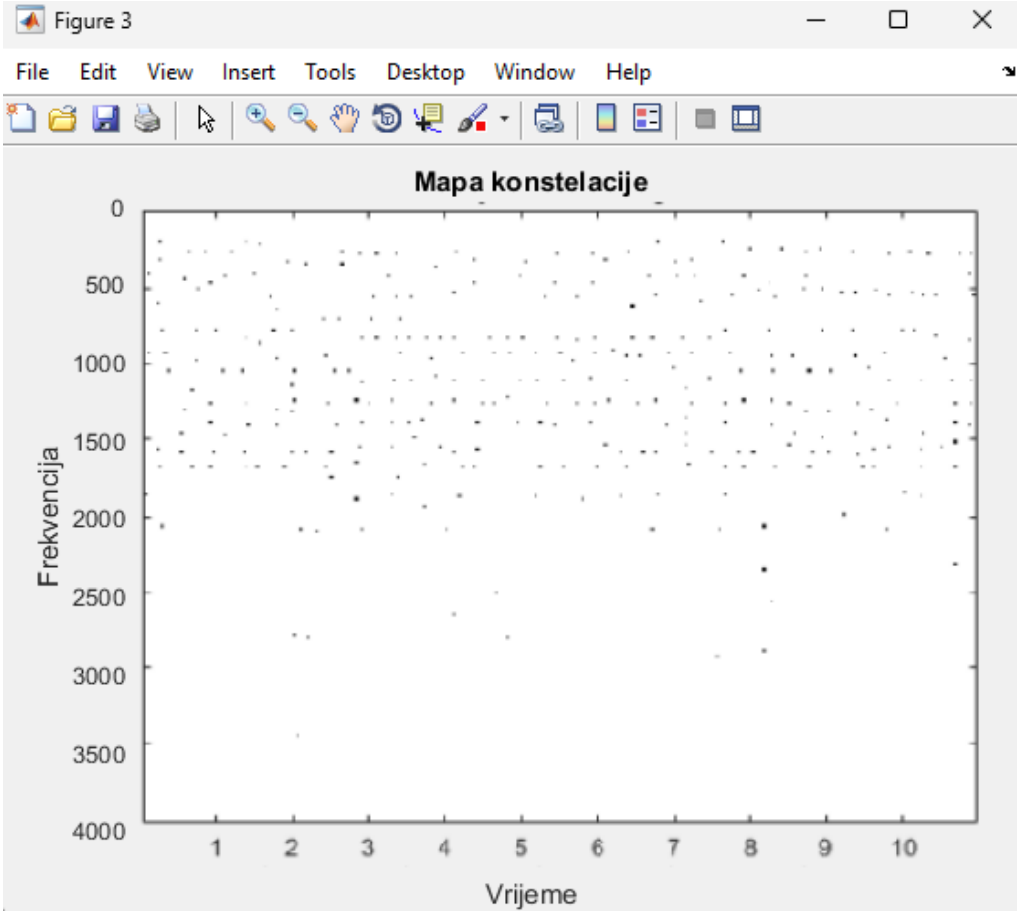
Figure 3
File Edit View Insert Tools Desktop Window Help
Mapa konstelacije
Frekvencija
0
500
1000
1500
2000
2500
3000
3500
4000
1 2 3 4 5 6 7 8 9 10
Vrijeme

Kod:

```
dt_l=0.05;
dt_u=0.4;
df=200;
out=3;
[I,J]=find(P);
features=[];
for il=1:length(I)
t1_indx=J(il);
f1_inx=I(il);
t1=T(t1_indx);
f1=F(f1_inx);
count=0;
prag_znacajke=[];
for jl=1:length(I)
t2_indx=J(jl);
f2_indx=I(jl);
t2=T(t2_indx);
f2_indx=I(jl);
t2=T(t2_indx);
f2=F(f2_indx);
if(t2>t1 +dt_l && t2<t1+dt_u && f2>f1-df &&
f2<f1+df)
prag_znacajke=[prag_znacajke;[f1_inx f2_indx
t1_indx t1_indx t1_indx-t2_indx]];
end
end
if ~isempty(prag_znacajke)
sortrows(prag_znacajke,4);
if size(prag_znacajke,1)>out
prag_znacajke=prag_znacajke(1:out,:);
end
znacajke=[znacajke;prag_znacajke(:,1:4)];
for j=1:size(prag_znacajke,1)
znacajka=prag_znacajke(j,:);
line([T(znacajka(3)),T(znacajka(5))],[F(znac
ajka(1)),F(znacajka(2))]);
end
end
end
```

Graf:

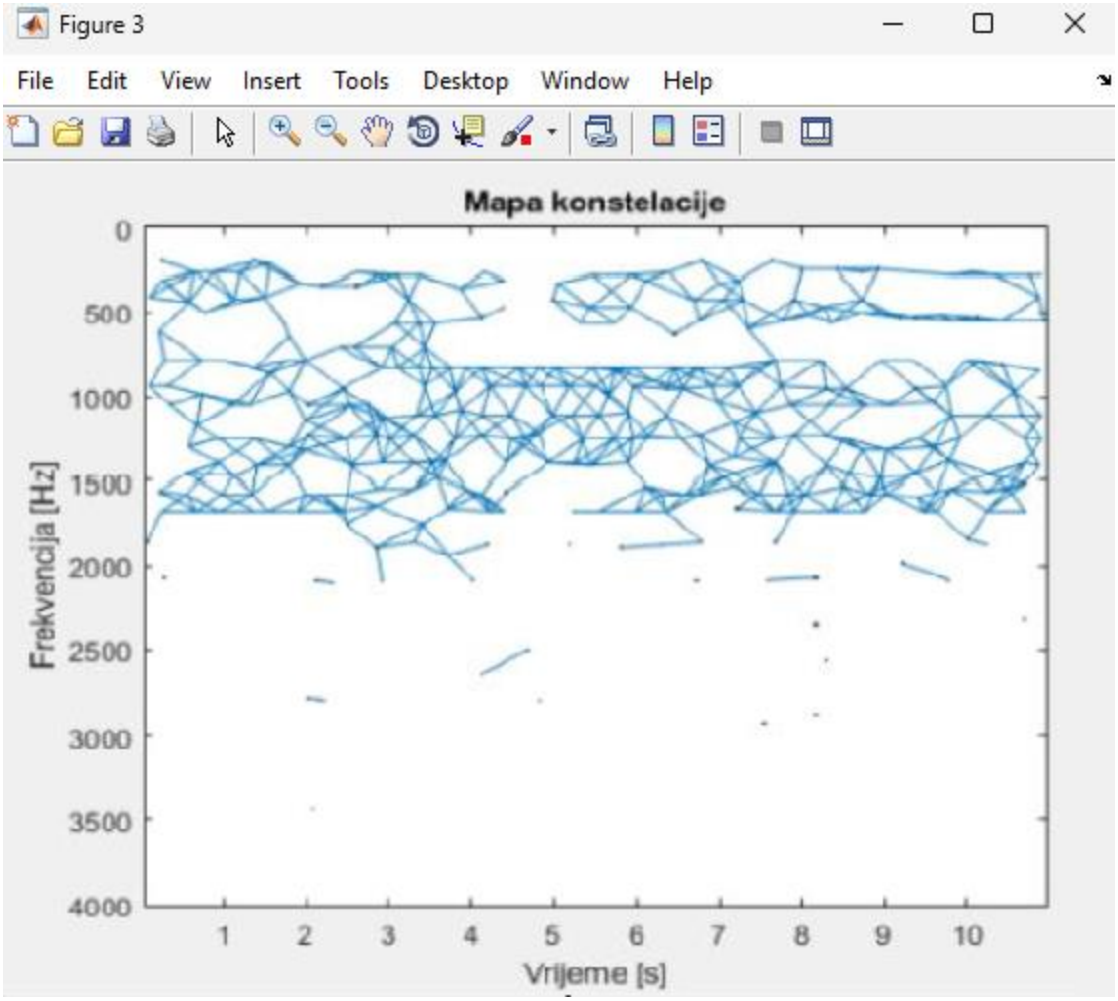
Figure 3
File Edit View Insert Tools Desktop Window Help
Mapa konstelacije
Frekvencija [Hz]
0
500
1000
1500
2000
2500
3000
3500
4000
1 2 3 4 5 6 7 8 9 10
Vrijeme [s]